

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 29

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. In High-Performance Computing (HPC), the Transpose operation is often used between row and column operations. Explain why it is often faster or easier for a processor to communicate with neighbors in the same row rather than across columns. How does performing a transposition improve spatial locality during the sorting process? If you have a system where each process is assigned to exactly one row, describe the communication pattern required to perform a column-wise sort without using a transpose operation.
2. You are given the following unsorted input array: [3, 0, 3, 2, 1, 1].
 - A. Create the initial count array where the index represents the value of the elements.
 - B. Perform the prefix sum transformation on your count array. What does the value at each index in this transformed array represent?
 - C. To maintain stability, the algorithm iterates through the original input array from right to left. Demonstrate the first two placements (the elements 1 and 1 at the end of the input) and show how the count array is updated after each placement.

- **Welcome: Sean**

- [LogOut](#)

